Proof by Mathematical Induction

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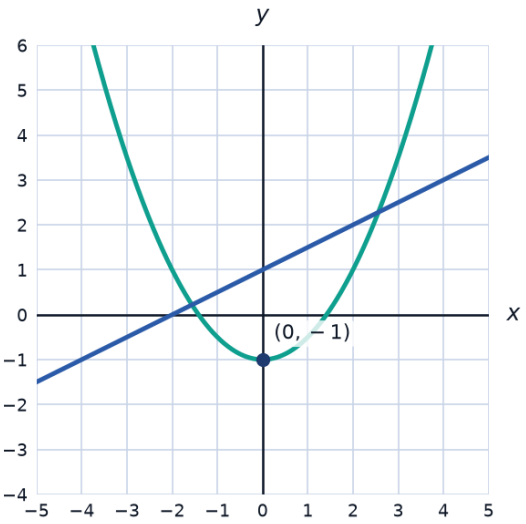
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: The three steps of induction

State the three steps required for a valid proof by mathematical induction.

2 [Foundational] • Target: Proving a summation formula

Prove by induction that $1 + 2 + 3 + \cdots + n = n(n + 1)/2$ for all $n \geq 1$.



3 [Foundational] • Target: Proving a summation formula

Prove by induction that $1^2 + 2^2 + 3^2 + \cdots + n^2 = n(n + 1)(2n + 1)/6$ for all $n \geq 1$.

4 [Foundational] • Target: Proving divisibility results

Prove by induction that $7^n - 1$ is divisible by 6 for all $n \geq 1$.

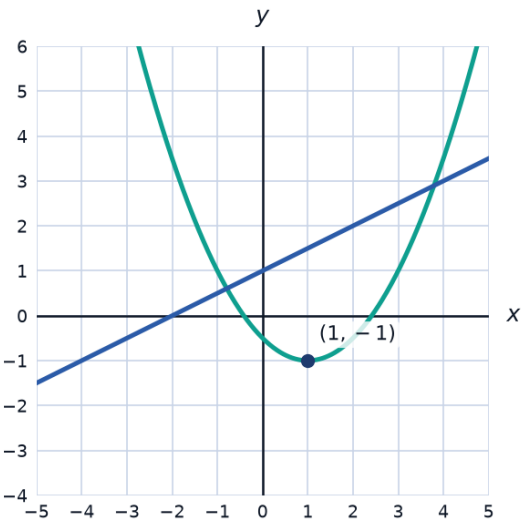
5 [Foundational] • Target: Induction with inequalities

Prove by induction that $2^n > n^2$ for all integers $n \geq 5$.

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6 [Foundational] • Target: Induction in calculus: derivatives of a product

Prove by induction that $\frac{d^n}{dx^n}(e^{ax}) = a^n e^{ax}$ for all positive integers n.



7 [Foundational] • Target: Induction with a recursively defined sequence

A sequence is defined by $u_1 = 3$ and $u_{n+1} = 2u_n - 1$. Prove by induction that $u_n = 2^n + 1$ for all $n \geq 1$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Proving a geometric-series-style summation

Prove by induction that the sum $3^0 + 3^1 + \cdots + 3^{n-1}$ equals $(3^n - 1)/2$ for all $n \geq 1$.

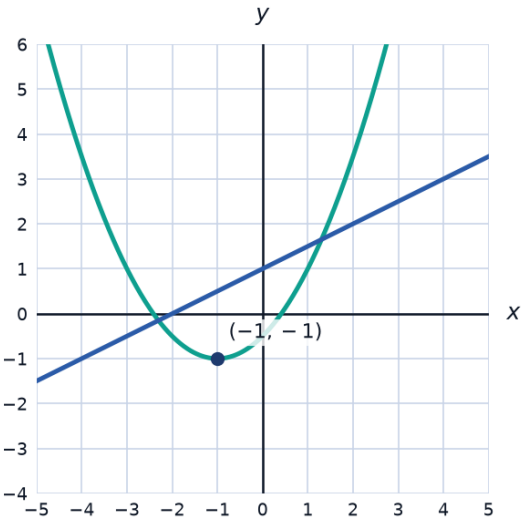
9 [Intermediate] • Target: Proving a geometric-series-style summation

Prove by induction that the sum $1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n!$ equals $(n+1)! - 1$ for all $n \geq 1$.

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10 [Intermediate] • Target: Proving an inequality involving factorials

Prove by induction that $n! > 2^n$ for all integers $n \geq 4$.



11 [Intermediate] • Target: Proving a divisibility result with two variables

Prove by induction that $n^3 + 2n$ is divisible by 3 for all positive integers $n \geq 1$.

12 [Intermediate] • Target: Proving a divisibility result with two variables

Prove by induction that $9^n - 1$ is divisible by 8 for all positive integers $n \geq 1$, and explain briefly why a similar argument would show $a^n - 1$ is divisible by $a - 1$ for any integer $a > 1$.

13 [Intermediate] • Target: Induction with a recurrence relation for a sequence of sums

A sequence satisfies $S_1 = 2$ and $S_{n+1} = S_n + 6n + 2$ for $n \geq 1$. Prove by induction that $S_n = 3n^2 - n$ for all $n \geq 1$.

14 [Intermediate] • Target: Induction with a strengthened hypothesis

Prove by induction that $11^n - 4^n$ is divisible by 7 for all positive integers $n \geq 1$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

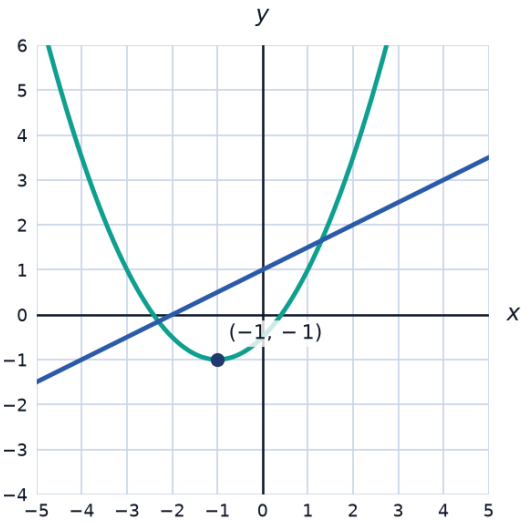
15 [Challenging] • Target: Induction to prove a formula for the sum of cubes • Cross-Lesson: 6.01 Matrices (AI HL)

Prove by induction that $1^3 + 2^3 + 3^3 + \cdots + n^3 = [n(n+1)/2]^2$ for all $n \geq 1$.

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16 [Challenging] • Target: Induction to prove a formula for the sum of cubes • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Prove by induction that $5^n + 2 \cdot 3^{n-1} + 1$ is divisible by 8 for all positive integers $n \geq 1$.



17 [Challenging] • Target: Induction to prove a formula for a weighted geometric sum • Cross-Lesson: 6.03 Markov Chains (AI HL)

Prove by induction that $1 \cdot 2^1 + 2 \cdot 2^2 + 3 \cdot 2^3 + \cdots + n \cdot 2^n = (n-1)2^{n+1} + 2$ for all $n \geq 1$.

18 [Challenging] • Target: A closing check: identifying an invalid induction argument • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

A student claims to prove that all horses are the same color using induction on the number of horses n in a group: the base case $n = 1$ is trivially true (one horse is trivially "the same color" as itself), and the student then argues that if any k horses are the same color, then any $k + 1$ horses must also be the same color. Explain precisely where this argument breaks down, and state why a correct induction proof must avoid this flaw.

19 [Challenging] • Target: Proving a sum-of-products formula by induction • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Prove by induction that $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \cdots + n(n+1) = n(n+1)(n+2)/3$ for all $n \geq 1$.

20 [Challenging] • Target: Proving an exponential inequality by induction • Cross-Lesson: 1.04 Partial Fractions

Prove by induction that $2^n > n$ for all integers $n \geq 1$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The Principle & Architecture of Induction

In a proof by mathematical induction that a proposition $P(n)$ is true for all integers $n \geq 1$, what are the essential logical components?

a. Base case: prove $P(1)$ is true; Inductive step: prove $P(k) \Rightarrow P(k+1)$; Conclusion by PMI

b. Verify $P(n)$ for $n = 1, 2, 3, 4, 5$ and deduce the pattern

c. Assume $P(n)$ is true for all n and differentiate both sides

d. Prove $P(k+1) \Rightarrow P(k)$ starting from an arbitrary large integer

22 [Foundational] • Target: The Principle & Architecture of Induction

If the base case of an induction proof is verified at $n = 4$, for which set of integers is $P(n)$ established (assuming the inductive step holds for all $k \geq 4$)?

a. $n \in \{4, 5, 6, \ldots\}$

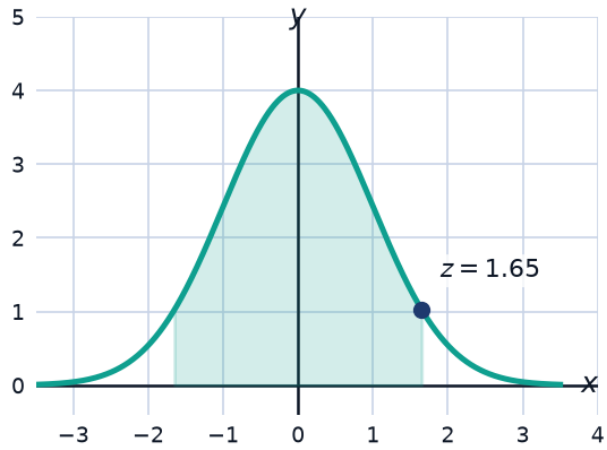
b. All integers $n \in \mathbb{Z}$

c. $n \in \{1, 2, 3, 4\}$

d. $n \in \mathbb{R}$ with $n \geq 4$

23 [Foundational] • Target: The Principle & Architecture of Induction

In proving $\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$, what is the algebraic expression for the sum up to $k+1$ using the induction hypothesis?



a. $\frac{k(k+1)}{2} + (k+1)$

b. $\frac{(k+1)(k+2)}{2} + k$

c. $\frac{k(k+1)}{2} \times (k+1)$

d. $\frac{k(k-1)}{2} + (k+1)$

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24 [Foundational] • Target: Summation and Series Proofs

When proving by induction that $\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$, factoring $(k+1)$ from $\frac{k(k+1)(2k+1)}{6} + (k+1)^2$ yields:

a. $\frac{k+1}{6}[2k^2+7k+6] = \frac{(k+1)(k+2)(2k+3)}{6}$

b. $\frac{k+1}{6}[2k^2+5k+6]$

c. $\frac{(k+1)^2(2k+1)}{6}$

d. $\frac{k+1}{6}[k(2k+1)+1]$

25 [Foundational] • Target: Summation and Series Proofs

For the proposition $\sum_{r=1}^{n}(2r-1) = n^2$, verify the inductive step by simplifying $k^2 + (2(k+1)-1)$.

a. $(k+1)^2$

b. k^2+2k

c. k^2+2k+3

d. $(k+2)^2$

26 [Foundational] • Target: Summation and Series Proofs

Consider the geometric series $\sum_{r=1}^{n} 2^{r-1} = 2^n - 1$. What is the base case check for $n = 1$?

a. $\text{LHS} = 2^0 = 1$, $\text{RHS} = 2^1 - 1 = 1$; LHS = RHS

b. $\text{LHS} = 2^1 = 2$, $\text{RHS} = 2^1 - 1 = 1$

c. $\text{LHS} = 0$, $\text{RHS} = 0$

d. $\text{LHS} = 1$, $\text{RHS} = 2$

27 [Foundational] • Target: Divisibility and Inequality Proofs

To prove $5^n - 1$ is divisible by 4 for $n \geq 1$, express $5^{k+1} - 1$ in terms of $(5^k - 1)$.

a. $5(5^k-1)+4$

b. $5(5^k-1)-4$

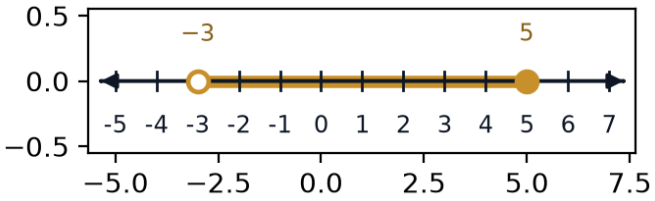
c. $4(5^k-1)+5$

d. $(5^k-1)+5$

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28 [Foundational] • Target: Divisibility and Inequality Proofs

In proving $2^n > n^2$ for $n \geq 5$, assuming $2^k > k^2$, which inequality must be justified to show $2^{k+1} > (k+1)^2$?



a. $2k^2 \geq (k+1)^2 \Leftrightarrow k^2 - 2k - 1 \geq 0$, which holds for $k \geq 5$

b. $k^2 > 2k + 1$ for all $k \in \mathbb{R}$

c. $2^k > 2k$

d. $(k+1)^2 > k^2 + 1$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Divisibility and Inequality Proofs

To prove $3^{2n} - 1$ is divisible by 8, evaluate the base case $n = 1$.

a. $3^2 - 1 = 8$, which is divisible by 8

b. $3^1 - 1 = 2$

c. $3^0 - 1 = 0$

d. $3^2 - 1 = 9$

30 [Intermediate] • Target: Calculus and Matrix Applications

Prove by induction that $\frac{d^n}{dx^n}(xe^x) = (x+n)e^x$. What is the inductive step derivative $\frac{d}{dx}[(x+k)e^x]$?

a. $1 \cdot e^x + (x+k)e^x = (x+k+1)e^x$

b. $(x+k)e^x + e^x = (x+k)e^x$

c. $(x+1)e^x + ke^x$

d. $(x+k+1)xe^x$

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31 [Intermediate] • Target: Calculus and Matrix Applications

If $A = [[1, 1], [0, 1]]$, prove by induction that $A^n = [[1, n], [0, 1]]$. What is the matrix product $A^k \cdot A$?

a. $[[1, k], [0, 1]][[1, 1], [0, 1]] = [[1, k+1], [0, 1]]$

b. $[[1, k], [0, 1]][[1, 1], [0, 1]] = [[1, 2k], [0, 1]]$

c. $[[1, k], [0, 1]][[1, 1], [0, 1]] = [[1, k^2], [0, 1]]$

d. $[[1, 1], [0, 1]]$

32 [Intermediate] • Target: Calculus and Matrix Applications

A sequence is defined by $u_1 = 5$ and $u_{n+1} = 2u_n - 3$. Prove by induction that $u_n = 2^{n+1} + 3$. Assuming $u_k = 2^{k+1} + 3$, what is u_{k+1}?

a. $2(2^{k+1} + 3) - 3 = 2^{k+2} + 6 - 3 = 2^{k+2} + 3$

b. $2(2^{k+1}) - 3 = 2^{k+2} - 3$

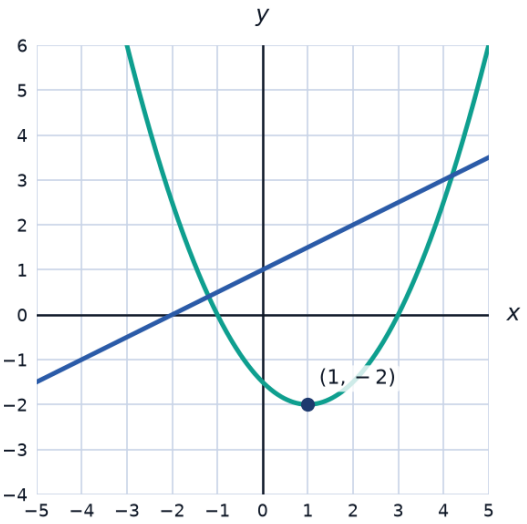
c. $2^{k+1} + 6$

d. $2^{k+2} + 6$

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33 [Intermediate] • Target: Induction to prove a formula for a weighted geometric sum

In proving by induction that $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$, assuming the result is true for $n = k$, what must be added to $\frac{k(k+1)}{2}$ to obtain the sum for $n = k+1$?



a. k

b. $k+2$

c. $k+1$

d. $2k+1$

34 [Intermediate] • Target: A closing check: identifying an invalid induction argument

Using the first three terms of the Maclaurin series for e^{2x}, approximate $e^{2 \times \frac{1}{5}}$ as a fraction.

a. $\frac{59}{50}$

b. $\frac{69}{50}$

c. $\frac{42}{25}$

d. $\frac{37}{25}$

35 [Intermediate] • Target: Proving a sum-of-products formula by induction

Using the first three terms of the Maclaurin series for e^{2x}, approximate $e^{2 \times \frac{1}{4}}$ as a fraction.

a. $\frac{13}{8}$

b. $\frac{57}{40}$

c. $\frac{53}{40}$

d. $\frac{77}{40}$

36 [Intermediate] • Target: Proving an exponential inequality by induction

Using the first three terms of the Maclaurin series for e^{x}, approximate $e^{1 \times \frac{1}{4}}$ as a fraction.

a. $\frac{173}{160}$

b. $\frac{41}{32}$

c. $\frac{253}{160}$

d. $\frac{237}{160}$

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37 [Intermediate] • Target: Proving a divisibility result by induction

Using the first three terms of the Maclaurin series for e^x, approximate $e^{1 \times \frac{1}{5}}$ as a fraction.

a. $\frac{51}{50}$

b. $\frac{38}{25}$

c. $\frac{71}{50}$

d. $\frac{61}{50}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Proving a closed form for a recursive sequence • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Using the first three terms of the Maclaurin series for e^x, approximate $e^{1 \times \frac{1}{2}}$ as a fraction.

a. $\frac{73}{40}$

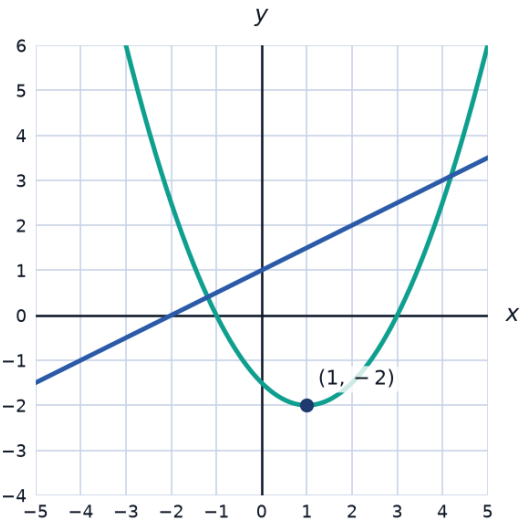
b. $\frac{13}{8}$

c. $\frac{57}{40}$

d. $\frac{61}{40}$

39 [Challenging] • Target: A second exponential inequality by induction • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

For $x^3 - 3x^2 - 36x + 108 = 0$, find the product of the three roots.



a. -73

b. -140

c. -134

d. -108

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40 [Challenging] • Target: A third inequality by induction • Cross-Lesson: 1.04 Partial Fractions

For $x^3 - 7x^2 - 14x + 120 = 0$, find the product of the three roots.

a. -128

b. -90

c. -156

d. -120

41 [Challenging] • Target: Induction in combinatorics: counting subsets • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

For $x^3 - 19x - 30 = 0$, find the product of the three roots.

a. 30

b. 43

c. 27

d. 20

42 [Challenging] • Target: Proving the sum of the first n odd numbers • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Using the first three terms of the Maclaurin series for e^{2x}, approximate $e^{2 \times \frac{1}{2}}$ as a fraction.

a. $\frac{11}{5}$

b. $\frac{5}{2}$

c. $\frac{12}{5}$

d. $\frac{14}{5}$

43 [Challenging] • Target: The Principle & Architecture of Induction • Cross-Lesson: 3.01 3D Vectors & the Cross Product

For $x^3 - 7x^2 + 10x = 0$, find the product of the three roots.

a. -1

b. -4

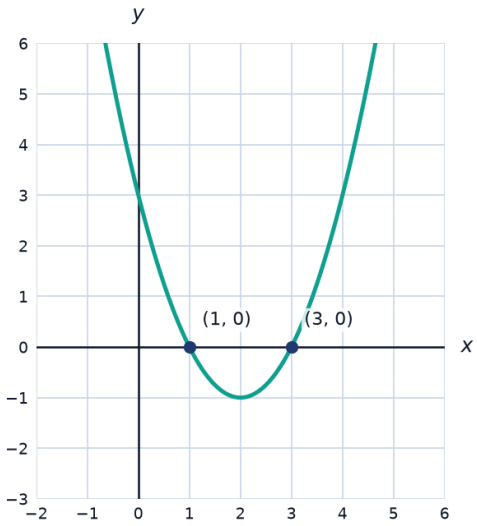
c. 0

d. -3

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44 [Challenging] • Target: Summation and Series Proofs • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. Determine the sum of the x-intercepts.



a. 4

b. 3

c. 1

d. -3

45 [Challenging] • Target: Divisibility and Inequality Proofs • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Product of roots of $2x^2 - 5x + 3 = 0$:

a. $\frac{3}{2}$

b. $\frac{5}{2}$

c. $-\frac{3}{2}$

d. 3

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